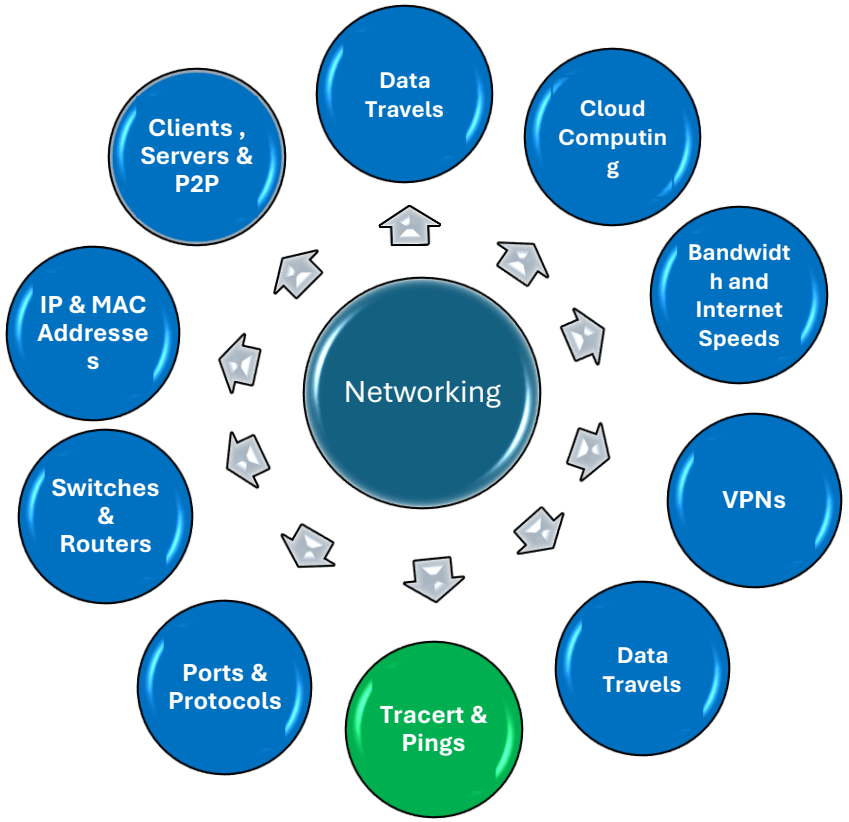
Networking
Data Travels
Cloud Computin g
Bandwidt h and Internet Speeds
VPNs
Data Travels
Tracert & Pings
Ports & Protocols
Switches & Routers
IP & MAC Addresse s
Clients , Servers & P2P

Ping & Traceroute Tools

```
C:\Users\CCENT>ping www.ipv6tb.edu

Pinging www.ipv6tb.edu [fdd2:8a70:f46:2::2] with 32 bytes of data:
Reply from fdd2:8a70:f46:2::2: time<1ms
Reply from fdd2:8a70:f46:2::2: time<1ms
Reply from fdd2:8a70:f46:2::2: time<1ms
Reply from fdd2:8a70:f46:2::2: time<1ms

Ping statistics for fdd2:8a70:f46:2::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\CCENT>tracert www.ipv6tb.edu

Tracing route to www.ipv6tb.edu [fdd2:8a70:f46:2::2]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  fdd2:8a70:f46:1::1
  2    <1 ms    <1 ms    <1 ms  fdd2:8a70:f46::2
  3    <1 ms    <1 ms    <1 ms  fdd2:8a70:f46:2::2

Trace complete.

C:\Users\CCENT>
```

Topic: *Network Diagnostic Tools – Ping & Traceroute*

Objective: Learn how to test and trace network connections using simple commands.

By the end of this lesson, you will know how to:

- Check if a device or website is reachable.
- Measure how fast data travels.
- Trace the path your data takes across the internet.

What Is Data Transmission?

When you send a message, open a website, or stream a video,
data travels across many routers until it reaches its destination.
Sometimes data is lost or delayed — we can test that using **Ping** and **Traceroute**.

What Is Ping?

Ping = Packet Internet Groper
It tests whether your computer can communicate with another device (host or website).

In simple terms:
Ping asks: “Hello, are you there?
”The target replies: “Yes, I’m here!”

🖥️ Example:
ping google.com
✅ If it replies → connection works
❌ “Request timed out” → no reply or delay

What Ping Tells You

When you ping, you get information like:

Field	Meaning
Packets Sent/Received	How many messages were sent and answered
Lost Packets	Missing replies = connection problem
Time (ms)	How long it took for data to go and return (latency)
IP Address	Real address of the website

Example result:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
Approximate round trip times: 32ms, 33ms, 31ms, 32ms

Common Ping Results

Result	Meaning
Reply from [IP]	The host is reachable
Request timed out	Connection lost or blocked
Destination unreachable	Network path not found
Ping statistics	Summary of all attempts

Tip:
Use ping -t website.com to keep it running continuously (until you press Ctrl + C).

Ping Example in Real Life

Command:
{ ping labcyber.com }
Output:
Reply from 35.208.192.32: bytes=32 time=28ms TTL=53

✅ 4 packets sent → 4 packets received → Connection OK 💡 Analogy:"You're saying hello 4 times — and the website replies 4 times."

What If It Says “Request Timed Out”?

Means:

- Temporary network issue
- Website/server down
- Firewall blocking ICMP packets
- Weak Wi-Fi or lost connection

Real-life analogy:

You call your friend, but they don’t pick up the phone 📞

What Is Traceroute (Tracert)?

Traceroute shows *the exact path* that your data takes from your computer to a destination.
Each stop along the way is called a **hop** (usually a router or server).

Windows command:
tracert facebook.com

Mac/Linux command:
traceroute google.com

How Traceroute Works

Traceroute sends small packets and increases the **Time-to-Live (TTL)** value each time.
Every router on the path replies with its **IP address and response time**.

Hop	IP Address	Time (ms)	Location
1	192.168.1.1	2 ms	Home Router
2	10.14.0.1	8 ms	ISP Network
5	216.239.63.141	82 ms	Google Server (USA)

📍 Each hop = one step your data takes through the Internet.

Reading Traceroute Results

Column	Description
Hop Number	Order of routers
Response Time	Delay in milliseconds
IP Address / Hostname	Network device name
Request Timed Out	Device didn't reply or blocked ping

If it stops showing results → packet reached its final destination.

Ping vs Traceroute

Feature	Ping	Traceroute
Purpose	Test connectivity	Show the route
Output	Reply time and loss	List of routers/hops
Command	ping	tracert (Windows) / traceroute (Mac/Linux)
Example Use	"Is Google reachable?"	"Where is the delay happening?"

Real-World Example

Problem: A company's website loads slowly.
IT Admin runs:

```
ping company.com
tracert company.com
```

- Ping shows high response time → network delay.
- Traceroute shows hop #8 taking 400 ms → problem found at the ISP router.

✅ Admin contacts ISP → issue fixed.

Quick Recap Quiz

1. What is the main purpose of the "ping" command?

A) To test the route between two computers B) To check if a device or website is reachable C) To block network connections D) To restart a router

2. Which protocol does "ping" use to test connectivity?

A) TCP B) ICMP C) FTP D) HTTP

3. When you see "Request Timed Out" after using the ping command, what does it mean?

A) The target computer replied too fast B) The target didn't respond or blocked ping requests C) The computer is in sleep mode D) The packet was too large

4. What does "ms" represent in ping results?

A) Memory size B) Milliseconds (round-trip time) C) Message speed D) Machine state

5. What is the main purpose of the "traceroute" (or tracert) command?

A) To send files over the network B) To show the path data takes from your device to a destination C) To download data faster D) To encrypt messages